Classical Encryption Techniques

Applied Cryptography

Learning Objectives, Examples, and Security Analysis

Learning Objectives

- Understand classical encryption techniques
- Analyze cryptanalytic attacks
- Study substitution and transposition ciphers
- Compare security of classical ciphers

Classical Encryption – Introduction

- Developed before modern computers
- Based on substitution and permutation
- Primary goal: confidentiality

Cryptanalytic Attacks

- Ciphertext-only attack
- Known-plaintext attack
- Chosen-plaintext attack
- Chosen-ciphertext attack

Security Criteria

- Size of key space
- Resistance to known attacks
- Confusion and diffusion
- Efficiency and practicality

Caesar Cipher

- Monoalphabetic substitution cipher
- Key = shift value (0–25)
- Encryption: $C = (P + k) \bmod 26$
- Easily broken by brute force

Caesar Cipher – Example

- Plaintext: ATTACK
- Key: k = 3
- Ciphertext: DWWDFN
- Only 25 possible keys

Hill Cipher

- Based on linear algebra
- Plaintext blocks as vectors
- Encryption: $C = KP \bmod 26$
- Key is an invertible matrix

Hill Cipher – Security

- Provides good diffusion
- Vulnerable to known-plaintext attacks
- Requires matrix inverse modulo 26

Vigenère Cipher

- Polyalphabetic substitution cipher
- Uses repeating keyword
- Reduces frequency patterns
- Encryption uses multiple Caesar shifts

Vigenère Cipher – Example

- Plaintext: ATTACKATDAWN
- Key: LEMON
- Ciphertext: LXFOPVEFRNHR
- Broken using Kasiski examination

Vernam Cipher

- Uses binary addition (XOR)
- Key length equals plaintext length
- Foundation of One-Time Pad

One-Time Pad (OTP)

- Key is truly random
- Key used only once
- Provides perfect secrecy
- Impractical key distribution

Comparison of Substitution Ciphers

- Caesar: very weak
- Playfair: moderate security
- Hill: strong but breakable
- Vigenère: stronger, not secure
- OTP: theoretically unbreakable

Security Comparison Table (Summary)

- Monoalphabetic → weakest
- Polyalphabetic → better security
- Random key + OTP → perfect secrecy
- Key management is critical

Transposition Cipher

- Rearranges character positions
- Does not change letter identities
- Preserves frequency distribution
- Examples: Rail Fence, Columnar

Transposition Cipher – Example

- Plaintext: MEETATNOON
- Rail Fence (2 rails)
- Ciphertext: MTAONETNO
- Often combined with substitution

Lecture Summary

- Classical ciphers form the basis of modern cryptography
- Security relies on key secrecy
- Most classical ciphers are vulnerable
- OTP achieves perfect secrecy theoretically

Quick Quiz

- Which cipher provides perfect secrecy?
- Which attack breaks Hill cipher?
- Why is Caesar cipher insecure?